

Rima Fathallah Software Engineering Graduate Student

✉ rimafathallah.9@gmail.com ☎ +216 56 479 019 📄 Rima Fathallah 🔗 Rima Fathallah

SUMMARY

A Computer Science graduate specialized in AI-driven solutions and intelligent systems, with over 3 years of practical experience in **Python**, **machine learning**, and **work automation**. Skilled in developing data-driven applications, optimizing analytical processes, and integrating intelligent models into real-world systems. Currently seeking an opportunity to advance my research capabilities and strengthen my professional expertise in AI and data science.

SKILLS

Programming Languages:

Python, R, C++, JavaScript, C#, SQL

Machine Learning & Data Science:

Scikit-learn, XGBoost, Random Forest, Pandas, NumPy, SciPy, Feature Engineering, Model Evaluation

Computer Vision & Research:

OpenCV, Image Processing, Object Detection, Image Classification, Experimental Design, Ablation Studies, Reproducible Research

Development & Tools:

Git, Docker, Flask, REST APIs, ReactJS, MySQL, PostgreSQL, LaTeX, Linux

Deep Learning & Generative AI:

PyTorch, TensorFlow, CNNs, RNNs/LSTMs, Transformers, Hugging Face, LLMs, Fine-tuning, AI Agents

PROFESSIONAL EXPERIENCE

AI Developer Engineer (Full Time) 🌐, Exceed Your Limits Marketing

Mar 2026 – Present | Germany, Remote

- Develop an AI-powered website that transforms user prompts into functional websites and automates code generation.
- Develop and optimize course, webinar, and payment workflows; integrate REST APIs and DB systems with 100K+ records.

Technologies: Python, AI/LLMs, C#, .NET, React, TypeScript, REST APIs, SQL, Docker, Git/GitLab

Product: <https://eyl-educate.com/> 🌐

Researcher (MITACS GRI'25 Intern) 🌐, Concordia University

Jan 2025 – Jul 2025 | Montreal, Canada

- Developed, calibrated, and validated ML models and prepared and analyzed datasets to support the research.
- Designed an automated platform, prototyped solutions, evaluated results, and documented findings for research teams.

Technologies: Python, Machine Learning, Data Analysis, Pandas, Scikit-learn, SQL, Jupyter, Git, XGBoost, LSTM

Junior Instructor (Part-Time) 🌐, Go My Code

Sep 2023 – Apr 2025 | Sousse, Tunisia

- Mentored students across AI, game development, and creative technologies, achieving a 95% project completion rate.
- Delivered hands-on training in Generative AI, Unity, Scratch, and Adobe, improving student performance by 30%.

Technologies: Generative AI, Unity, Scratch, Adobe Illustrator

Junior Artificial Intelligence Engineer (Internship) 🌐, Ozeol.com

Jun 2024 – Sep 2024 | Sousse, Tunisia

- Designed optimized data pipelines, improving processing efficiency by 27% and analyzing 5K+ records for realtime reports.
- Built data collection and validation workflows that reduced data inconsistencies by 13% while supporting secure platforms.

Technologies: Python, FastAPI, SQL, Pandas, Web Scraping, Data Analysis, Data Pipelines, Real-Time Reporting

Junior Software Engineer (Internship) 🌐, Draexlmaier

Aug 2023 – Oct 2023 | Sousse, Tunisia

- Designed an anonymous employee feedback platform that increased employee engagement by 25%.
- Built real-time dashboards and analytics that improved management's response to workplace issues by 20%.

Technologies: Power Apps, Power BI, Data Analytics, Data Visualization, SQL

Automation Engineer (Part-Time), Elit school of languages Sousse

Jun 2022 – Oct 2022 | Sousse, Tunisia

- Automated data processing and management workflows, reducing processing time and improving productivity by 30%.
- Developed an automated certification system with real-time data updates, improving user satisfaction by 20%.

Technologies: Python, Pandas, NumPy, Data Automation, Data Processing, Real-Time Systems

Junior Web Developer (Internship) 🌐, BillCom Consulting

Jul 2022 – Sep 2022 | Sousse, Tunisia

- Designed and optimized a scalable data platform, improving database performance by 18%.
- Automated Linux-based deployment and maintenance workflows, improving platform stability and reducing manual task.

Technologies: Python, MySQL, SQL, Linux, Database Optimization, Data Engineering, Automation

CERTIFICATES

Fundamentals of Deep Learning 🌐

NVIDIA Deep Learning Institute

Building Transformer-Based Natural Language Processing (NLP and LLM) Applications 🌐

NVIDIA Deep Learning Institute

Applications of AI for Predictive Maintenance 🌐

NVIDIA Deep Learning Institute

AI-900: Azure AI Fundamentals 🌐

Microsoft Certified

Introduction to Security Principles in Cloud Computing 🌐

Google Cloud Security - Coursera

RESEARCH AND DEVELOPMENT

AI-Driven Modeling of Neuroplasticity Trajectories in Pediatric Cerebral Palsy Aug 2026 – Present
Leading an ongoing AI research project in collaboration with Sahloul University Hospital and Hached University Hospital.

PROJECTS

- NEXOVA** *🔗, Startup Idea* Sep 2024 – present
AI-powered platform that analyzes and optimizes energy consumption to reduce electricity usage and improve efficiency.
Technologies: Python, Machine Learning, Data Analysis, Pandas, NumPy, Scikit-learn
- Autonomous Data Analysis Agent**, *Personal Development Project* Jul 2026 – Sep 2026
Developed an AI agent that autonomously cleans, analyzes, visualizes, detects anomalies, and reports based on requests.
Technologies: Python, Pandas, NumPy, Scikit-learn, LLMs/SLMs, Function Calling, LangGraph, FastAPI, Plotly, Docker
- Multimodal AI System for Patient Risk Assessment**, *Mid-Term Project* Apr 2026 – Jun 2026
Developed a multimodal AI system combining clinical data, medical images, and patient history to predict patient risk.
Technologies: Python, CNN, Vision Transformers (ViT), XGBoost, Transformers, SHAP, Grad-CAM, Pandas, Scikit-learn
- Explainable Multilingual Misinformation Detection**, *Evaluation Project* Feb 2026 – Apr 2026
Developed a multilingual NLP system for misinformation detection across languages.
Technologies: Python, NLP, TF-IDF, LSTM, BERT, AraBERT, Multilingual Transformers, Explainable AI, Scikit-learn, Hugging Face
- ToolMind**, *School Project* Dec 2025 – Feb 2026
Developed an intelligent AI agent that plans tasks, selects tools/APIs, and generates responses using structured tool routing.
Technologies: Python, SLMs (Llama), Hugging Face, LangGraph, LangChain, FastAPI, REST APIs, Scikit-learn, Docker, MLflow
- DEXTRA**, *Congress Challenge* Sep 2025 – Dec 2025
AI-powered smart gloves enabling real-time bidirectional translation between sign language and speech using embedded ML.
Technologies: Python, Machine Learning, Embedded AI, Computer Vision, Sensor Data, IoT
- Talynk** *🔗, Personal Development Project* Aug 2025 – Nov 2025
Developed an AI-powered platform for CV generation and interview preparation using NLP and machine learning.
Technologies: Python, NLP, Machine Learning, LLMs, Text Processing, Scikit-learn

EDUCATION

- Engineering Degree in Data Science**, *Ecole Polytechnique de Sousse - Evening courses* *🔗* Sep 2025 – Present | Sousse, Tunisia
Main Modules: MLOps, AIoT, MERN, NLP, C, AI, CyberSec
- Bachelor's degree in Computer Science (Licence)** Sep 2023 – Jul 2025 | Sousse, Tunisia
École supérieure des sciences et de la technologie de Hammam Sousse(ESSTHS) *🔗*
Main Modules: Algorithms, Industrial Programming, Advanced OO Programming, ISTQB, web programming, IoT, AI & ML
- Bachelor's degree in Mathematics (Licence)** Sep 2021 – Jun 2023 | Sousse, Tunisia
École supérieure des sciences et de la technologie de Hammam Sousse(ESSTHS) *🔗*
Main Modules: Statistics with R, Probabilities, Operational research, Queuing Theory, Algebra & Analysis

LANGUAGES

English	French	German	Arabic
DET - C1	DELFI - B2	ELIT - A1	Native

AWARDS

- AI NIGHT Challenge** *🔗, ARSII, Second Place in the 2026 Edition, Fourth place in the 2024 Edition* 05 May 2026
- Herotopia Challenge** *🔗, IEEE TSYP 13, First Place* 24 Dec 2025

VOLUNTEERING

- UNESCO Chaire EDE X IEEE**, *Program Co-Lead* Jan 2026 – Dec 2026
- Led 30+ ambassadors to develop monthly sustainable projects, delivering measurable impact across Tunisia.
- IEEE Tunisia Section - SIGHT Group**, *Project Coordinator* Jan 2025 – Dec 2026
- Coordinates several Sustainable Projects that align with SDGs in Tunisia, Sustainable Technology, ThyroCare, Bled el Khir...
- IEEEExtreme 18.0 - Region 8** *🔗, Co-Lead* Sep 2023 – Oct 2024
- Led a team of 90+members across Africa, Europe, and ME to create world's largest CP competition with 10K+ participants.
- IEEE Tunisia Section - ESSTHS Student Branch** *🔗, Chair* Jan 2023 – Dec 2023
- Led the largest SB in Tunisia. winning 1st place in 8+ competitions. Organized TSYP 11 with 1,200+ participants and 15 challenges.